

Computer Science

Master of Science

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PROGRAM DESCRIPTION

The graduate program is designed to equip students with advanced knowledge in both computer science theory and application. This includes opportunities through funded research, seminars, real world project, and lab-work to acquire hands-on experience on a wide variety of state-of-the-art computer hardware and software systems.

The department offers a rigorous Master of Science degree in Computer Science, providing basic competence in the principal areas of computer science. On this foundation, the student has the opportunity to build expertise in one of a number of practical and theoretical subjects, working toward a deeper understanding under the guidance of a faculty member.

ADMISSION REQUIREMENTS

Please see the general graduate admission requirements [here](#). Specific requirements to this program are as follows:

1. Prospective students must have substantial background in computing, mathematics and science. The required computing background typically includes: competence in programming using C++/Java, discrete structures, data structures and algorithms, computer organization and architecture, programming language theory and operating systems. The required mathematics and science background include two courses in college calculus, a linear algebra course, and exposure to mathematical logic. A student must have a GPA of 3.00 in these, as well as an overall undergraduate GPA of 3.00 otherwise the student must take required undergraduate background courses as specified by the graduate advisory committee at the time of admission.
2. To be admitted to the master's program, the applicant must have the equivalent of an undergraduate degree in computer science from a regionally accredited college or university.

Specific requirements to the CMP-AMP Option are as follows:

1. Applicants must finish his/her junior year.
2. Applicants should hold at least a 3.5 cumulative GPA at the end of their junior year to be accepted into the CMP-AMP Graduate Program. GRE will be waived under (1) 3.7+ GPA or (2) 133+ Major Field Test score.
3. Students need to take three graduate courses in the senior year once (s)he is admitted to the AMP option.
4. For the security area, students need to take two graduate security courses and one graduate course once admitted.
5. Students can apply for B.S. once finished 120 credit hours at the undergraduate level.
6. After entering graduate school, the students will complete the rest of the eight graduate courses, i.e., three Fall courses, three Spring courses, two Summer courses.

DEGREE REQUIREMENTS

The program provides for thesis and non-thesis options.

Thesis Option

Students who choose to take the thesis option must complete 33 credit hours of coursework. The coursework consists of 18 hours of core courses and 9 hours of computer science (CS) electives. The master's research and thesis (6 hours) must be an original work that (1) offers a theoretical contribution to the field or (2) provides a new methodology or techniques for solving practical problems in the area of computer science.

For students in the CMP-AMP Option, typically, it takes four years for an undergraduate degree and two years for a graduate degree in computer science. In the AMP Option, during their fourth year, students will take nine credit hours of graduate core courses. The remaining courses (27 hours including six credits of thesis) will be taken in three semesters (Summer, Fall and Spring) in their last year. These students are committed to research projects under their advisors. The AMP Option is for thesis option only.

Non-thesis Option

Student who chooses the non-thesis option must complete 36 hours of coursework. The coursework consists of 18 hours of core courses and 18 hours of computer science (CS) electives. With this option, the student must pass the comprehensive exam within three attempts. The comprehensive examination will consist of questions from the knowledge units from each of the six core courses. A score of 75 or better is required to pass the comprehensive examination. Students also must complete all core courses prior to taking the comprehensive examination.

Core Courses

Every student must take all of the core courses to fulfill the breadth-first requirement. A minimum GPA of 3.00 must be earned in the core courses.

Elective Courses

A number of elective courses are provided for the student to master in interested field(s) of specialization.

(CMP) Computer Science – Thesis incl. AMP Option[#]

33 Credit Hours

MinGPA cumulative 3.0. MinGrade C*. Degree M.S.

CORE COURSES	
CS 511 Design & Anal of Algorithms	3
CS 521 Obj Oriented Prgming & Design	3
CS 531 Computer Architecture	3
CS 541 Operating System Principles	3
CS 551 Database Mgt Systems	3
CS 561 Software Engg Methodology	3
ELECTIVES	
CS 5xx	3
CS 5xx	3
CS 5xx	3